

Introduction

Grey squirrels (*Sciurus carolinensis*) and european rabbits (*Oryctolagus cuniculus*), preyed upon by red foxes (*Vulpes vulpes*) and raptors, coexist with these predators in Silwood park. Both squirrels, rabbits and foxes show some level of habituation to human presence. Anthropophony can both protect and disrupt prey ecology, depending on noise type and consistency [6, 4]. For foxes, which tend to thrive in suburban zone due to greater shelter and food opportunities [5] anthropogenic food sources can reduce home range size even when environmental conditions would otherwise favor larger foraging areas [3].

Habitat selection results from trade-offs between predation risk and resource availability. In a forest environment such as Silwood Park, food and shelter are easily accessible for squirrels and rabbits. We can therefore expect top-down mechanisms to be the key structuring drivers of habitat selection in prey species. Through the risk-disturbance hypothesis, [1] suggests that human disturbances - such as noise - and predation risk are both components of the landscape of fear. Using the Normalized Difference Soundscape Index (NDSI) as a proxy for disturbed area, I hypothesize that human presence in Silwood park indirectly pushes grey squirrels to invest acoustic ecotones, that is habitats with enough anthropogenic noise to deter or disturb avian predators [2], but not enough to overlap strongly with foxes' home range, concentrated in highly disturbed area.

Materials and methods

Data collection and processing Acoustic sensors (AudioMoth) and camera traps were deployed across Silwood park to monitor wildlife presence and soundscapes. Acoustic data were processed using *tuneR*, *soundecology* and *seewave* packages to extract the Normalized Difference Soundscape Index (NDSI), which quantifies the balance between anthropogenic and biological sounds. A +1 NDSI indicates a soundscape dominated by biophony, while a -1 indicates dominance by anthropogenic sounds. Camera trap images were analyzed to identify species presence. NDSI was calculated over 10-minutes intervals every hour in order to get a representative measure of the soundscape over the 16 days of recording while reducing data volume. Camera trap images were timestamped to match the acoustic data (ie. hourly intervals). Thereby, a good trade-off between resolution and variability in animal presence and soundscape conditions was achieved.

However, when focusing on our 3 species of interest, the data was cut out to only 3 sensors (Grid 45, 60 and 68) which recorded a sufficient number of detections over the study period (sensor 18 was discarded as it showed only one data point). We can see on Figure 1 and Table 1 that foxes are mainly detected in grid 68 corresponding to a suburban area, compared to the two other sensor locations situated in more forested areas, thereby confirming the intuition behind the hypothesis.

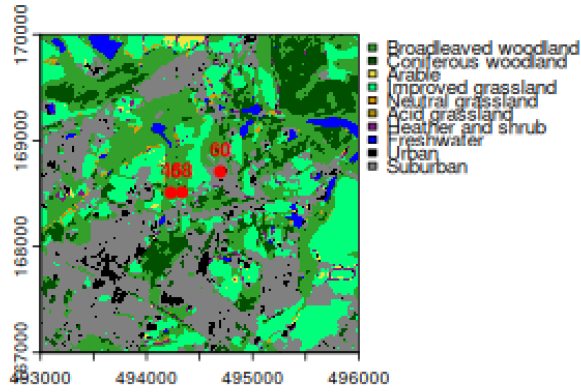


Figure 1: Location of the 3 selected sensors and their cover-type.

	species	grid	n
1	Oryctolagus_cuniculus	45	6
2	Oryctolagus_cuniculus	68	9
3	Sciurus_carolinensis	45	35
4	Sciurus_carolinensis	60	23
5	Sciurus_carolinensis	68	3
6	Vulpes_vulpes	60	1
7	Vulpes_vulpes	68	8

Table 1: Number of detections per species and sensor grid.

Data analysis After an exploratory vizualization of species density across the NDSI gradient, I plan to use an ANOVA test to show that mean NDSI value of rabbits, squirrels and foxes habitat significantly differ, thus supporting the hypothesis that squirrels and rabbits select habitat based on anthropogenic noise levels to avoid predation risk. A Tukey HSD post-hoc test will be performed to test pairwise difference between species.

Results

References

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